MILITARY STANDARD

MIL-STD-1320-201 A
(NAVY)

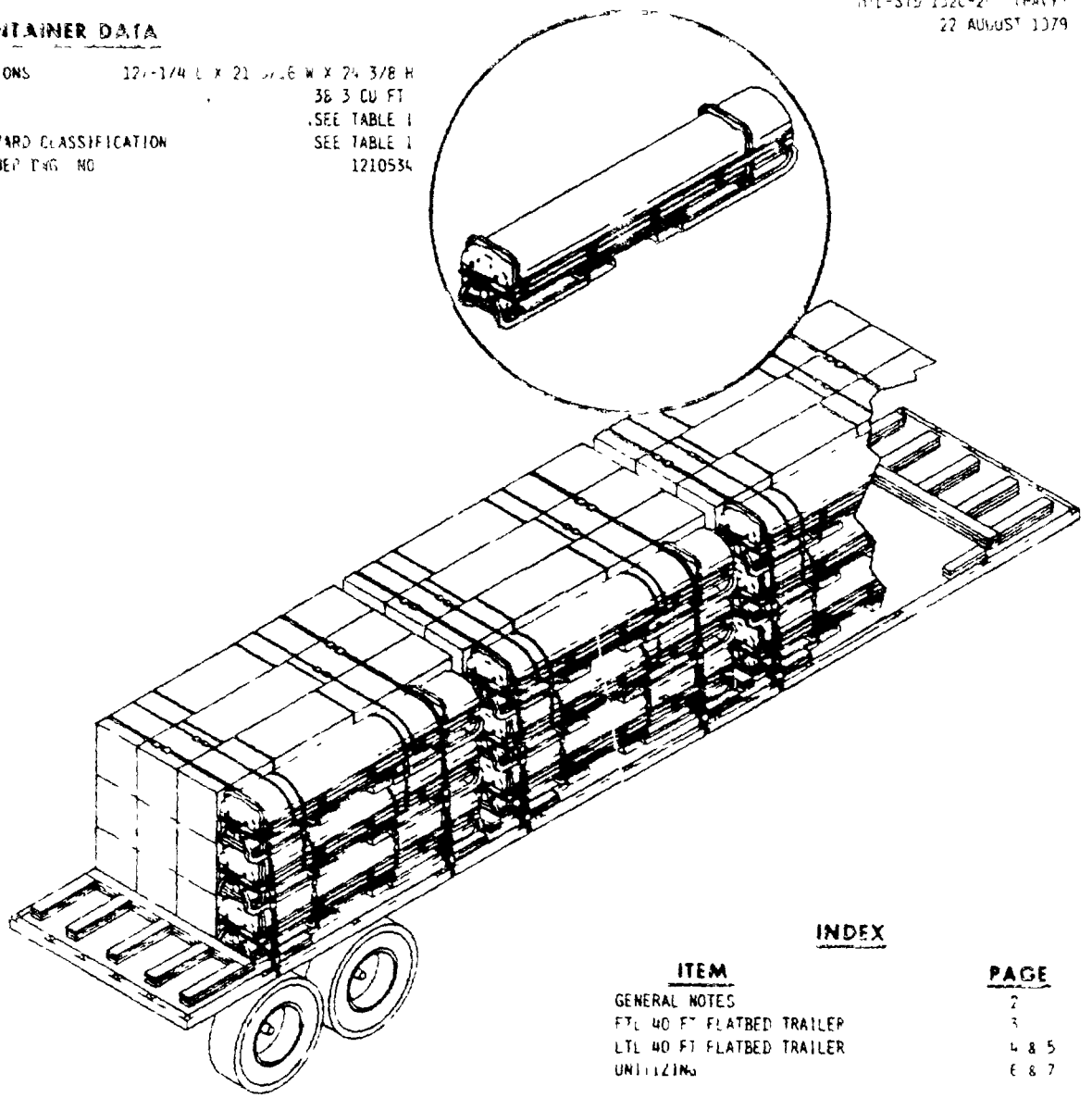
22 Aug 1980

SUPERSEDING
MIL-STD 1320-20[illegible] (NAVY)
22 AUGUST 1979

TRU[illegible] LOADING
TORPEDO MK 46 ASSEMBLIES
IN CONTAINER MK 535 MOD 0

CONTAINER DATA

DIMENSIONS	12[illegible]-1/4 L X 21 [illegible]/8 W X 24 3/8 H
CUBE	38.3 CU FT
WEIGHT	SEE TABLE I
DOT HAZARD CLASSIFICATION	SEE TABLE I
CONTAINER DWG NO	1210534



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 FT FLATBED TRAILER	3
LTL 40 FT FLATBED TRAILER	4 & 5
UNITIZING	6 & 7

NOTES

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

2 FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
A	ADDED SH 7 PROVIDING ALTERNATE UNITIZING	8/14/90	PK	[illegible]
		APPROVAL		

FSC 8140

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SIGNATURE [illegible] DATE

SIGNATURE SEASYSCOM DIRECTION DATE

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PAGE

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GENERAL NOTES

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR PRELOADING THE MK 46 TORPEDO ASSEMBLIES IN CONTAINER MK 535 MOD 0
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS THE TRAILER AXLE SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER) DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS
3. WHEN CONTAINERS ARE TWO OR THREE HIGH, THEY SHALL BE UNITIZED AS SHOWN ON PAGE 6

WARNING

WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED CONTAINERS EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING SPECIAL ATTENTION SHOULD BE GIVEN TO APPROPRIATE BACKUP OF OUTBOARD CONTAINERS CONTAINERS SHALL BE DEUNITIZED AFTER UNLOADING THE VEHICLE

NOTE FOR ALTERNATE METHOD OF UNITIZING SEE PAGE 7

4. EACH STACK OF TWO OR THREE HIGH CONTAINERS IS SECURED TOGETHER WITH TWO 1 1/4 X .035 CROSS STRAPS THE STRAPS PASS OVER THE COVERS OF THE CONTAINERS IN THE TOP LAYER AND RETURN PASSING UNDER THE BOTTOMS OF THE SAME CONTAINERS BINDING THE TOP LAYER TOGETHER TENSION STRAPS AND SEAL WITH TWO 1 1/4 SEALS DOUBLE CRIMPED OR ONE 1 1/4 SEAL DOUBLE NOTCHED
5. CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY ON A ONE TO ONE BASIS FOR TIEDOWNS
6. CHAINS, FITTINGS AND LOADBINDERS SHALL MEET THE REQUIREMENTS OF THE BASIC TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY) AND APPLIED AS SHOWN ON PAGE 5 OF THIS DOCUMENT
7. STEEL STRAP USED AS TIEDOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE BASIC TRUCKLOADING DOCUMENT MIL-STD- 1320 (NAVY) AND AS SHOWN ON PAGE 3 OF THIS DOCUMENT
8. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626 THE CHAINS FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH GOUGING BENT LINKS WEAR AND ANY OTHER NOTICEABLE DEFECTS THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO 22 OF DD FORM 626 ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILTY OT THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR -TRAILER WILL NOT EXCEED THESE LIMITATIONS
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO DUNNAGE IN AN ACCESSIBLE AREA AND WHEN REQUIRED (SEE TABLE I) ATTACH APPROPRIATE PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER
11. UNLESS OTHERWISE SPECIFIED ALL MATERIAL SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)
12. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY

FTL 40 FT TRAILER (FLATBED)

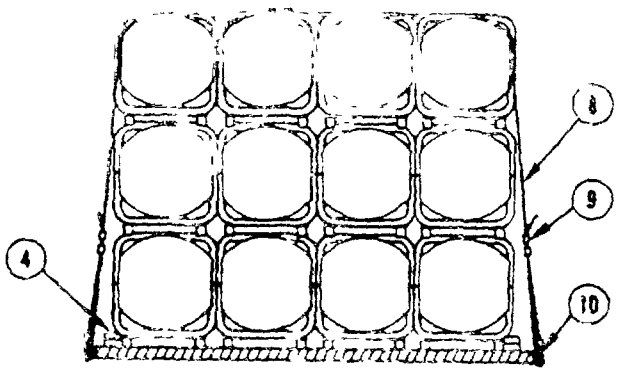
1. A FULL TRUCKLOAD FOR A 40 FT FLATBED TRAILER CONSISTS OF 36 CONTAINERS

TABLE I

* N/A - NOT APPLICABLE

DODIC/ NALC	ITEM		GROSS WT OF ONE CONTAINER (LBS (APPROX))	FTL WEIGHT (LBS. (APPROX))	DOT HAZARD CLASSIFICATION
T026	MAIN ASSEMBLY, TORPEDO MK 46 MOD 1	W/O FUEL	604	21,744	INERT
		W/FUEL	692	24,912	* N/A
T029	MAIN ASSEMBLY, TORPEDO MK 46 MOD 2	W/O FUEL	604	21,744	INERT
		W/FUEL	692	24,912	* N/A
T031	MAIN ASSEMBLY, TORPEDO MK 46 MOD 1 (PHASE 2)	W/O FUEL	604	21,744	INERT
		W/FUEL	692	24,912	* N/A
T044	DUMMY TORPEDO, MK 46 MOD 1 (WOOD)		788	28,368	INERT
T045	DUMMY TORPEDO, MK 46 MOD 1 (STEEL)		788	28,368	INERT
1510	TORPEDO MK 46 MOD 1, PHASE 2, WARSHOT COMPLETE ASSEMBLY, W/O LAUNCH ASSEMBLY		786	28 296	A
1511	TORPEDO MK 46 MOD 1 PHASE 2 WARSHOT COMPLETE ASSEMBLY TUBE LAUNCHED		788	28 368	A
1512	TORPEDO MK 46 MOD 1 WARSHOT COMPLETE ASSEMBLY TUBE LAUNCHED		788	28 368	A
1527	TORPEDO MK 46 MOD 1 EXERCISE COMPLETE ASSEMBLY TUBE LAUNCHED	W/LONG TANK	788	28 368	* N/A
		W/SHORT TANK	744	26 784	
1528	TORPEDO MK 46 MOD 1 EXERCISE [illegible] TUBE LAUNCHED		[illegible]	[illegible]	
1543	TORPEDO MK 46 MOD 2 EXERCISE [illegible]	W/LONG TANK	788	28 368	* N/A
		W/SHORT TANK	[illegible]	[illegible]	

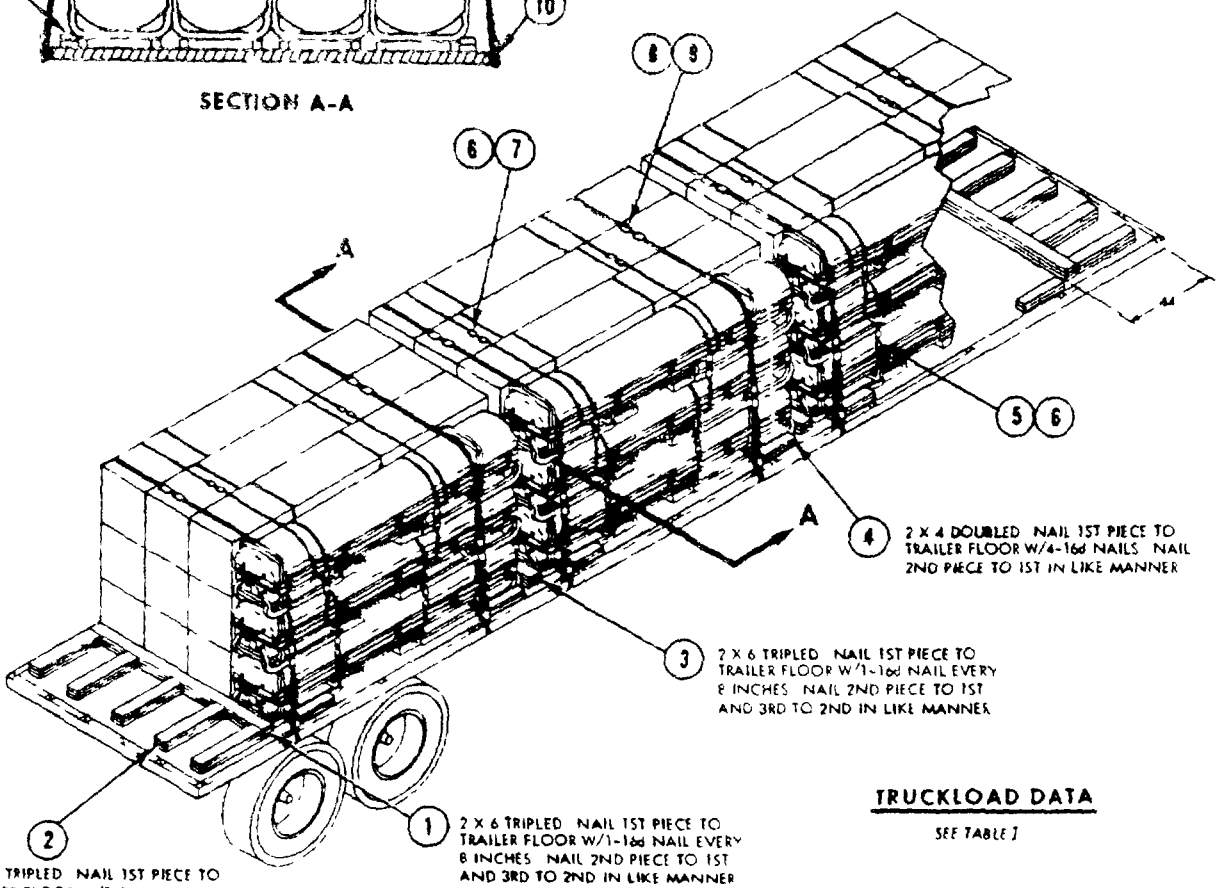
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SECTION A-A

NOTE

THIS FTL SHOWS THE PROPER USE OF 2 X .050 STRAPS AS TIEDOWNS CHAINS ARE AN AUTHORIZED ALTERNATE AND SHOULD BE APPLIED AS SHOWN ON PAGE 5



4 2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/4-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

3 2 X 6 TRIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER

2 2 X 6 TRIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/5-16d NAILS NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER

1 2 X 6 TRIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER

TRUCKLOAD DATA

SEE TABLE I

10	STAKE POCKET PAD (STRAP)	1 1/4 X .035 X 18	12	-		
9	SEAL	FOR 2 IN STRAP	48	-	-	
8	TIEDOWN STRAP	2 X .050 X 24 FT 6	6	-	-	-
7	CROSS STRAP	1 1/4 X .035 X 19 FT	6	-	-	-
6	SEAL	FOR 1 1/4 STRAP	108	-	-	-
5	UNITIZING STRAP	1 1/4 X .035 X 8FT 6	48	-	-	
4	SLEEPER	2 X 4 X 20	24	SEE FIELD NOTE		
3	INTERMEDIATE CROSSMEMBER	2 X 6 X 86	6	SEE FIELD NOTE		
2	BACKUP CLEAT	2 X 6 30	30	SEE FIELD NOTE		
1	END CROSSMEMBER	X X 86	6	SEE FIELD NOTE		

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	

(1) (1) MATERIAL & NAILING DATA

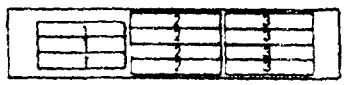
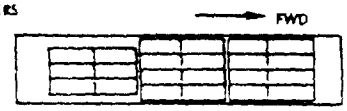
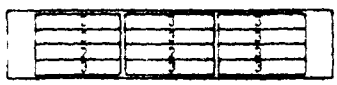
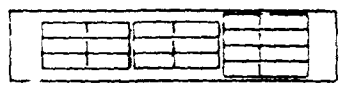
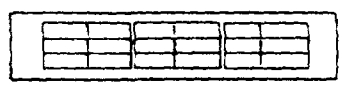
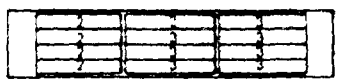
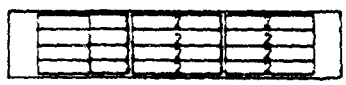
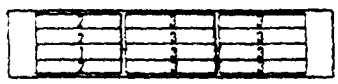
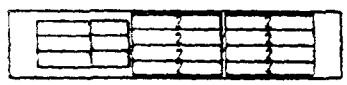
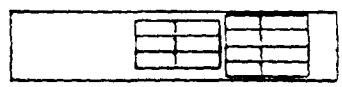
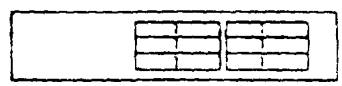
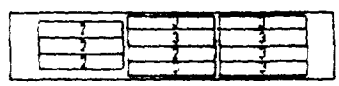
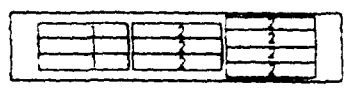
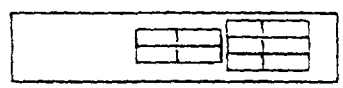
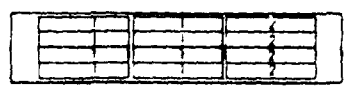
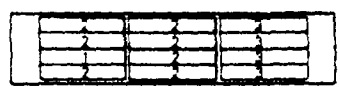
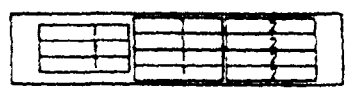
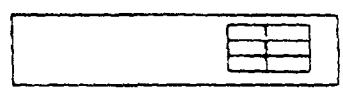
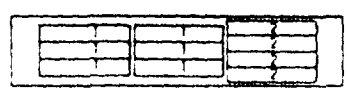
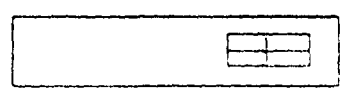
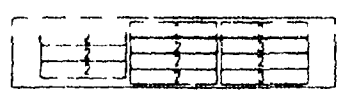
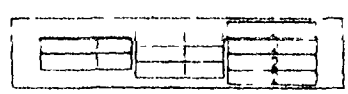
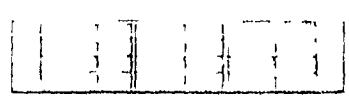
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LTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LTL, SELECT THE CORRECT LOAD PATTERN FROM TABLE II FOR THE NUMBER OF TORPEDOES TO BE SHIPPED WHEN A STACK DOES NOT HAVE A COMPLETE SECOND LAYER (E.G. LOAD) FOR 25, 27, 31, 32, 34 AND 35 CONTAINERS USE THE FILLER ASSEMBLY DETAIL B TO FILL THE VOID. THE ASSEMBLY SHALL BE STRAPPED TO THE CONTAINER ON WHICH IT RESTS USING THE PRINCIPLES SHOWN FOR UNITIZED CONTAINERS ON PAGE 6
2. BLOCK AND TIEDOWN ALL LTL'S USING THE PRINCIPLES SHOWN ON PAGES 3 & 5
4. THE LOAD SHOWN ON PAGE 5 IS FOR 25 TORPEDOES. SHIPMENTS ARE NOT LIMITED TO THIS QUANTITY. THIS LOAD IS SHOWN TO DEMONSTRATE THE PRINCIPLES OF BLOCKING AND TIEDOWN FOR LTL SHIPMENTS
5. THE LOAD PATTERNS SHOWN IN TABLE II MAY BE VARIED WHEN CIRCUMSTANCES DICTATE. AXLE WEIGHT LIMITATIONS SHALL BE OBSERVED. THE USE OF SHORTER/LONGER TRAILERS ARE AUTHORIZED PROVIDED THE BLOCKING AND TIEDOWNS ARE IN ACCORDANCE WITH THE PRINCIPLES OF THIS DOCUMENT. SHIPMENT MAY BE MADE USING "DOUBLES" WHEN "DOUBLES" ARE AUTHORIZED IN THE STATE(S) THROUGH WHICH THE SHIPMENT IS ROUTED

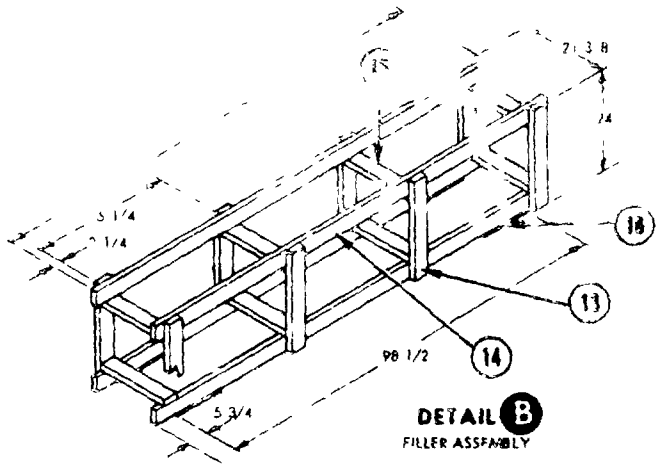
TABLE II

* LOAD PATTERNS

NO. OF CONTAINERS	FWD	NO. OF CONTAINERS	FWD	NO. OF CONTAINERS	FWD
35		23		11	
34		22		10	
33		21		9	
32		20		8	
31		19		7	
30		18		6	
29		17		5	
28		16		4	
27		15		3	
26		14		2	
25		13		1	
24		12			

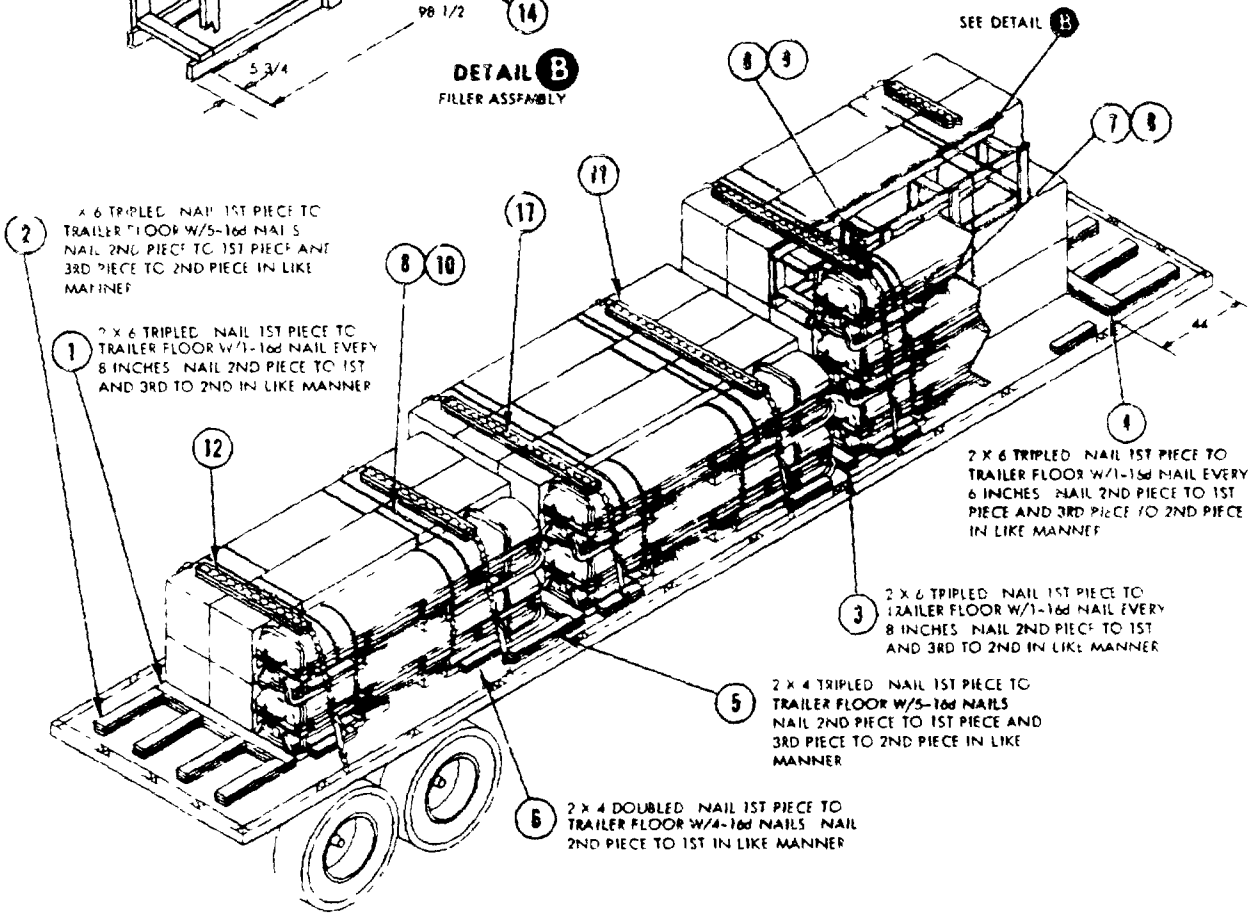
* [illegible] THE [illegible] OF THE LOAD. THE NUMBER IN THE BLOCK [illegible] THE [illegible] CONTAINERS IN THE PILE

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NOTE

THIS LTL SHOWS THE PROPER USE OF CHAINS AND LOADBINDERS. TWO INCH STRAP IS AN AUTHORIZED ALTERNATE AND SHOULD BE APPLIED AS SHOWN ON PAGE 3



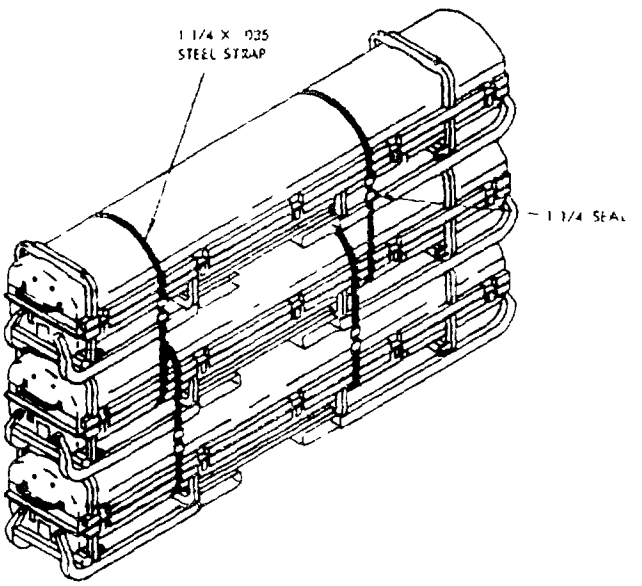
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TY	NAILS NUMBER	NAILS SIZE
17	CHAIN & LOADBINDER	5/16 OR 3/8	6		-	-
16	STOP (PLYWOOD)	1/2 X 1 1/2 X 12	4	14	3/JOINT	8d
15	STRUT	2 X 4 X 18 3/4	8	14	2/JOINT	10d
14	HORIZONTAL	2 X 4 X 110	4	SEE 13	-	-
13	VERTICAL	2 X 4 X 24	8	14	2/JOINT	10d
12	PROTECTOR BOARD	2 X 6 X 53	4	DOU-BLED	5	10d
11	PROTECTOR BOARD	2 X 6 X 74	4	DOU-BLED	-	10d
10	CROSS STRAP	1 1/4 X .035 X 15 FT 6	6		-	-
9	CROSS STRAP	1 1/4 X .035 X [illegible]	4			

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TY	NAILS NUMBER	NAILS SIZE
8	SEAL	FOR 1 1/4 STRAP	72			-
7	UNITIZING STRAP	1 1/4 X .035 X 8 FT 6	30		-	
6	SLEEPER	2 X 4 X 20	24	SEE FIELD NOTE		
5	INTERMEDIATE BACKUP CLEAT	2 X 4 X 30	6	SEE FIELD NOTE		
4	FWD CROSSMEMBER	2 X 6 X 86	-	SEE FIELD NOTE		
3	INTERMEDIATE CROSS MEMBER	2 X 6 X 86	6	SEE FIELD NOTE		
2	BACKUP CLEAT	2 X 6 X 30	2	SEE FIELD NOTE		
1	REAR CROSSMEMBER	2 X 6 X 64	2	SEE FIELD NOTE		

LTL 60 FT TRAILERS (25 TORPEDOES)

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CONTAINER UNITIZING



1. USING A FORKLIFT TRUCK STACK THE MIDDLE CONTAINER ON THE BOTTOM CONTAINER ORIENTING CONTAINERS AS SHOWN MAKING CERTAIN STACKING FEATURES ARE ENGAGED
2. STRAP THE MIDDLE CONTAINER TO THE BOTTOM CONTAINER WITH TWO 1-1/4 X .035 STEEL STRAPS AND SEAL WITH TWO DOUBLE NOTCHED SEALS OR ONE DOUBLE NOTCHED SEAL. THE STRAPS PASS OVER THE TOP OF MIDDLE CONTAINER AND UNDER THE BODY OF THE BOTTOM CONTAINER
3. STACK THE TOP CONTAINER ON TOP OF THE MIDDLE CONTAINER ORIENTING CONTAINERS AS SHOWN MAKING CERTAIN STACKING FEATURES ARE ENGAGED. STRAP TOP CONTAINER TO BOTTOM CONTAINER USING METHOD IN STEP TWO. (OMIT THIS STEP WHEN REQUIRED TO STACK CONTAINERS TWO HIGH.)

WARNING

WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED CONTAINERS, EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING. SPECIAL ATTENTION SHOULD BE GIVEN TO APPROPRIATE BACKUP OF OUTBOARD CONTAINERS. CONTAINERS SHALL BE DEUNITIZED AFTER UNLOADING THE VEHICLE.

NOTE: FOR ALTERNATE METHOD OF UNITIZING SEE PAGE 7

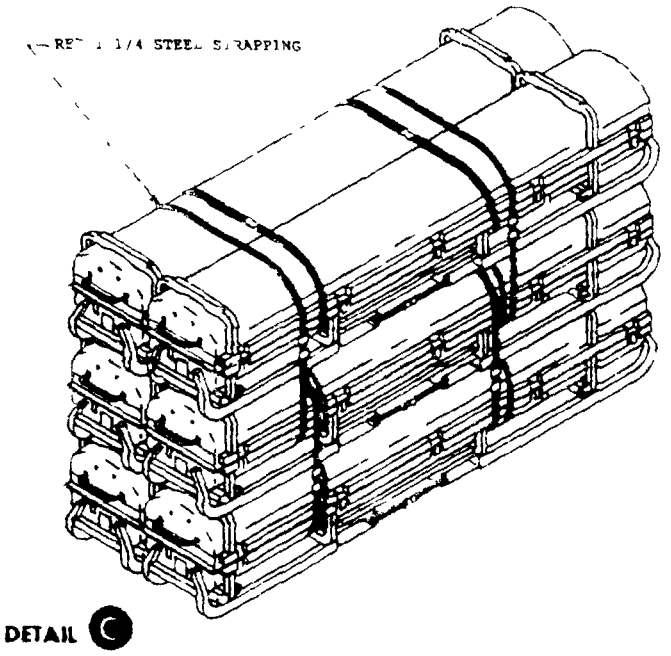
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ALTERNATE UNITIZING PROCEDURE

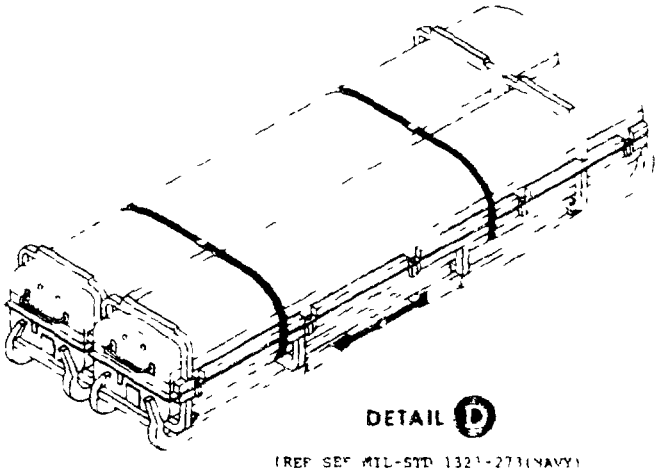
(1) A FORKLIFT TRUCK STACK TWO (2) OR THREE (3) UNIT LOADS OF DETAIL D (REQUIRED TOGETHER AS SHOWN IN DETAIL C. BE SURE TO ALIGN THE STACKING FEATURES. THE MIDDLE CONTAINERS ARE STRAPPED TO THE BOTTOM CONTAINERS AND THE TOP CONTAINERS TO THE MIDDLE CONTAINERS USING 1-1/4 X .035 STEEL STRAPPING, TENSIONED AND SEALED WITH TWO (2) DOUBLE NOTCHED SEALS. THE STRAPS PASS OVER THE SKIDS OF THE UPPER CONTAINER AND UNDER THE BODY OF THE LOWER CONTAINER.

WARNING

THE STRAPPING TOGETHER OF UNIT LOADS TO FORM A UNIT OF CONTAINERS TWO (2) OR THREE (3) HIGH IS AUTHORIZED FOR LOADING AND UNLOADING OF TRAILERS ONLY. ANY MOVEMENT OF THESE LOADS IS LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER.



DETAIL C



DETAIL D

(REF SEE MIL-STD 1321-273(NAVY))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R355

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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1320-201A - Truckloading - MK 46 Torpedo in Container MK 535

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
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